Aadit Gupta

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Education

University of Toronto **Sep. 2024 – Apr. 2029 (expected)**
Bachelor of Applied Science and Engineering in Computer Engineering + PEY Co-op *Toronto, ON*
- Relevant Courses: Engineering Strategies and Practices; Electrical Fundamentals; Calculus III; Computer Fundamentals (C/C++); Digital Systems (Verilog/FPGA Design)

Experiences

Machine Learning Researcher | University of Strathclyde - Glasgow, UK | GitHub Link **Jun. 2026 – Aug. 2026**
- Built an **end-to-end digital twin** that decides in real time whether an AED-delivery drone can safely beat an ambulance to a cardiac-arrest call across Greater Glasgow & Clyde
- Engineered an **A*** drone router over a population/wind cost grid with a **PyTorch** flight-time surrogate model
- Optimized drone-base placement with a multi-objective **NSGA-II genetic algorithm** (pymoo), raising reachable cardiac-arrest demand from **47% to 72%**
- Implemented a JARUS SORA 2.5 **aviation-risk engine** with a **per-flight go/no-go engine**, displayed using a Flask operator console featuring animated fleet simulation and coverage analytics.

Product Management Intern | Andersen UAE - Dubai, UAE **Jun. 2025 – Aug. 2025**
- Developed an **AI-powered E-Invoicing Assistant** that processed **client tax queries in under 5 seconds.**
- Built and deployed a **RESTful API** with **FastAPI** enabling seamless integration into client systems.
- Cleaned and Analyzed **10000+ data entries** using **Pandas** and stored them in a **ChromaDB Vector Database**.
- Optimized responses using **retrieval + caching techniques** from the Database **decreasing response times by 40%.**
- Engineered response pipelines using OpenAI's **GPT-4 LLM** through the **OpenAI API** reducing latency by **1.3 seconds**.

Camera Strap Redesign – Engineering Strategies | Communication Liaison - Toronto, CA **Jan. 2025 – Apr. 2025**
- Worked with a client to **design** a solution to improve camera straps by making it **more ergonomic.**
- **Responsible** for all communication between client and team in coordination with a Manager.
- Created a **Gantt Chart** and **2 Status Reports** to keep track of the team's progress.
- Prototyped **3 potential designs**, iterating based on client and user feedback to achieve an optimized solution.

Co-Curricular Activities & Projects

University of Toronto Autonomous Scale Racing | Hardware Team Member **Sep. 2025 - Present**
- Wrote code in **C++** that enabled **autonomous communication** between an Arduino and an Nvidia Jetson.
- Designed a custom part of a **PCB** on Altium that allowed communication between the Arduino Nano and the Jetson.
- Debugged **UART**-based communication by addressing timing issues and data size constraint of **32 Bytes** on the Jetson .

University of Toronto Aerospace Team | Software Team Member **Sep. 2025 - Present**
- Developed a **camera-based target localization system** using **Python and OpenCV** for autonomous missions.
- Implemented **minimum-jerk trajectory planning** in **MATLAB/Python**, focusing on smooth and stable flight paths.
- Containerized workflows using **Docker** to ensure reproducible development environments and consistent testing.

Streetview Monopoly Game Project **Jan. 2026 – Mar. 2026**
- Built a **full-stack multiplayer game** leveraging **Google Streetview** for location-based gameplay.
- Developed server-side logic for turn-based state management, player actions, and game rule enforcement.
- Integrated **Google Maps API** with environment-based configuration for secure **API key handling.**

FPGA Design Project **Sep. 2025 – Nov. 2025**
- Designed a rhythm game using the **DE1-SoC Board,** which incorporated a VGA Display, Keyboard and Speakers
- Implemented FSMs, Shift Registers and Counters in **Verilog** to implement game logic and output graphics on the VGA.

Technical and Professional Skills

Coding Languages: Python, C/C++, TypeScript, SQL
Frameworks & Libraries: React, Node.js, Django, Tailwind, FastAPI, SQLite, ChromaDB, OpenCV
Tools & Technologies: Git, Docker. MATLAB, LTSpice, Altium
Professional Skills: Teamwork, Communication, Problem Solving, Leadership